

UNIT IV

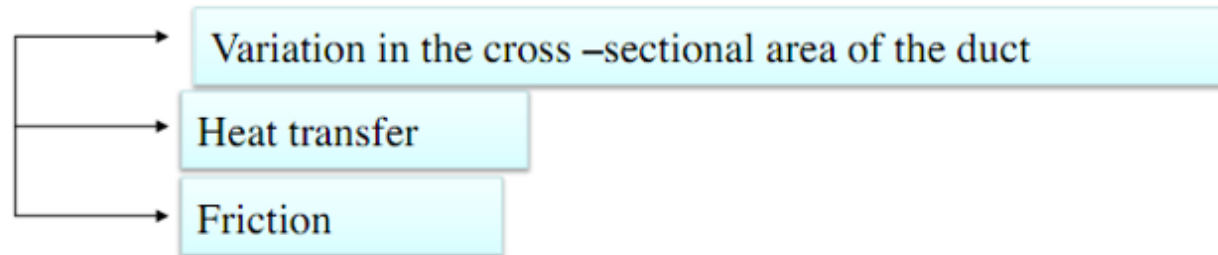
Fanno Flow : Flow with friction in constant area duct, Fanno lines, Fanno equation, Definition of friction constant, Friction loss, Effect of wall friction on flow properties, Local flow properties in terms of local, Mach number (No Derivation-Only Numericals)

Rayleigh Flow : Flow with heating or cooling in ducts, Governing equations, Slope of Rayleigh line, Entropy considerations. Maximum heat transfer (No Derivation-Only Numericals)

Simple frictional flow (Fanno Flow)

Adiabatic frictional flow in a constant-area duct

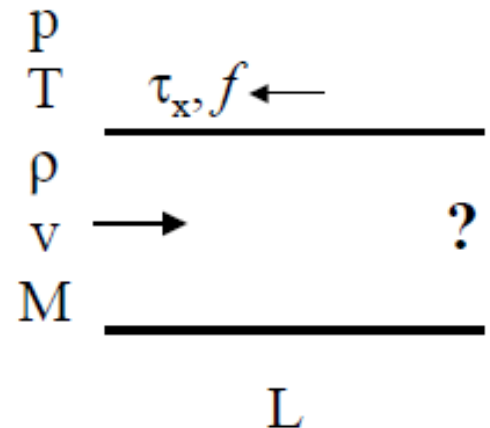
* The Flow of a compressible fluid in a duct is
Always accompanied by :-



The above parameters contribute to changes in the flow properties.

Fanno Flow

- Steady, 1-d, constant area, adiabatic flow with no external work but *with friction*

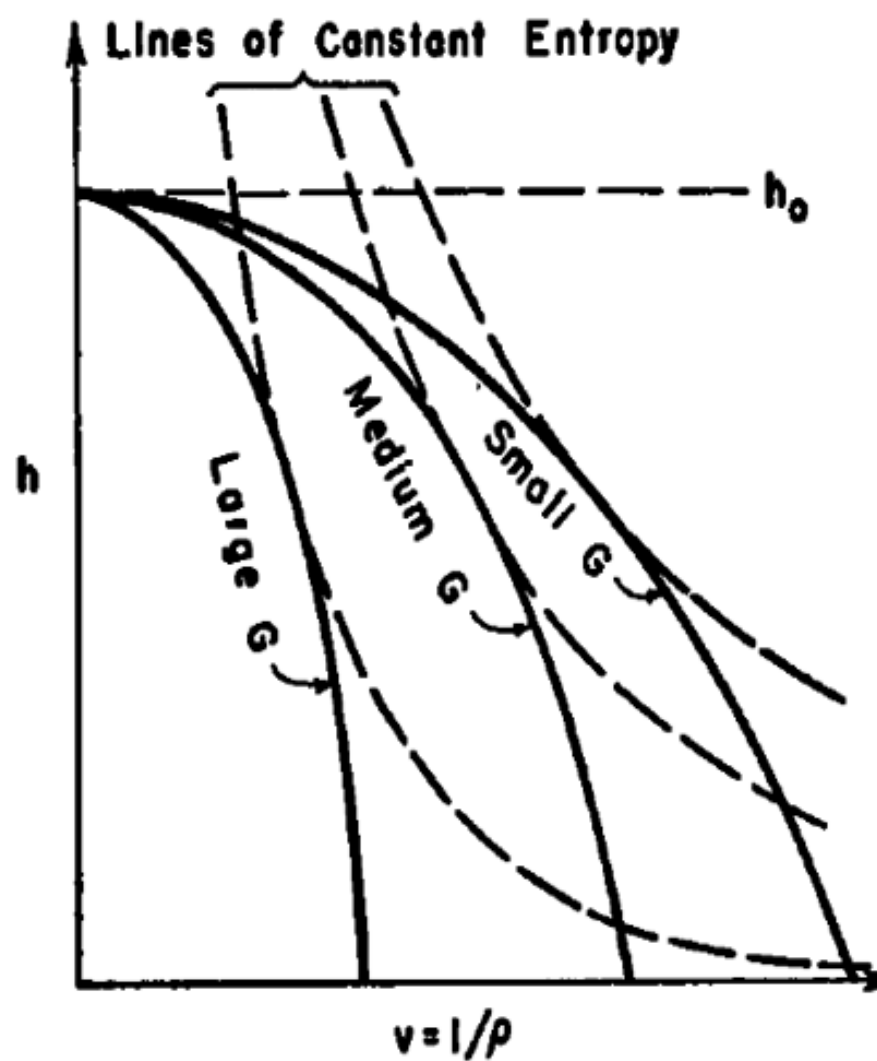


- Conserved quantities
 - since adiabatic, no work: **$h_o = \text{constant}$**
 - since $A = \text{const}$: mass flux = **$\rho v = \text{constant} \equiv G$**
 - combining: **$h_o = h + G^2 / 2\rho^2 = \text{constant}$**

As you change h ,
you change ρ (and v)
since G and h_o const.

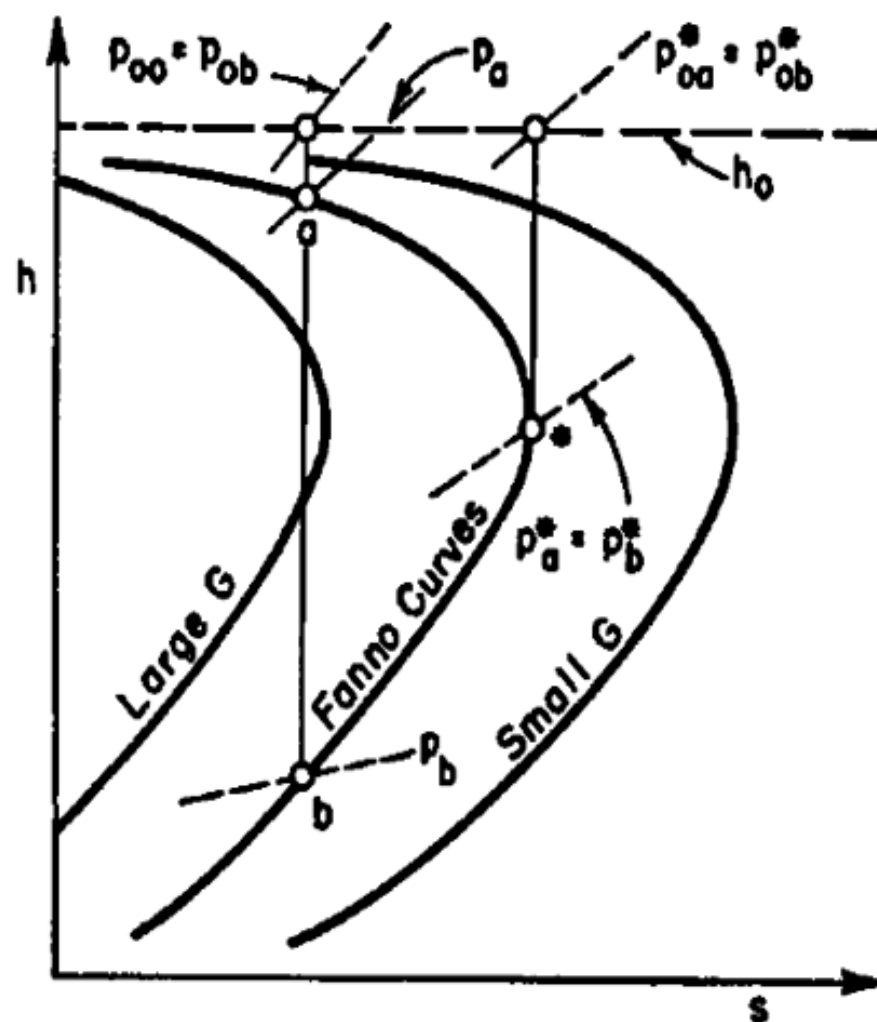
- On h - s diagram, can draw **Fanno Line**
 - line connecting points with same h_o and ρv

$$h = h_0 - \frac{G^2}{2\rho^2}$$



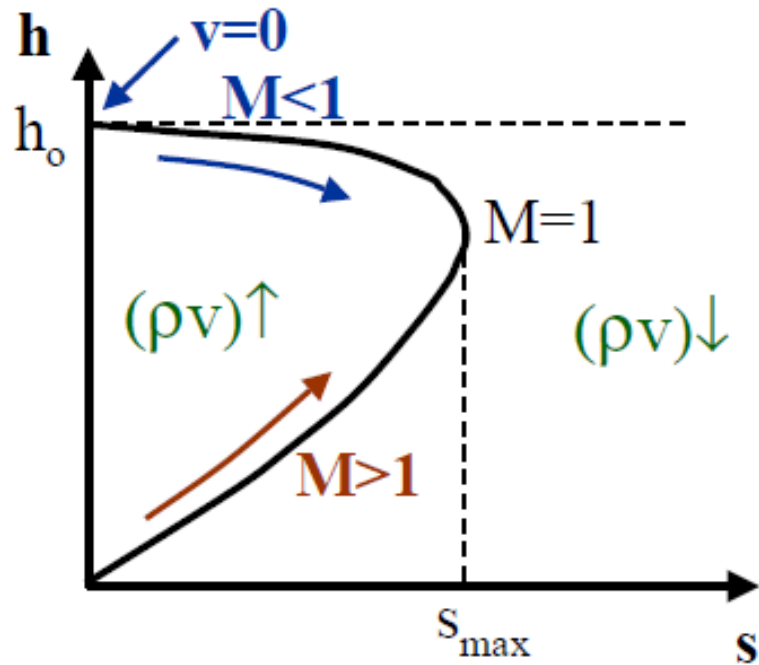
For a pure substance,

$$s = s(h, \rho)$$



Fanno Line

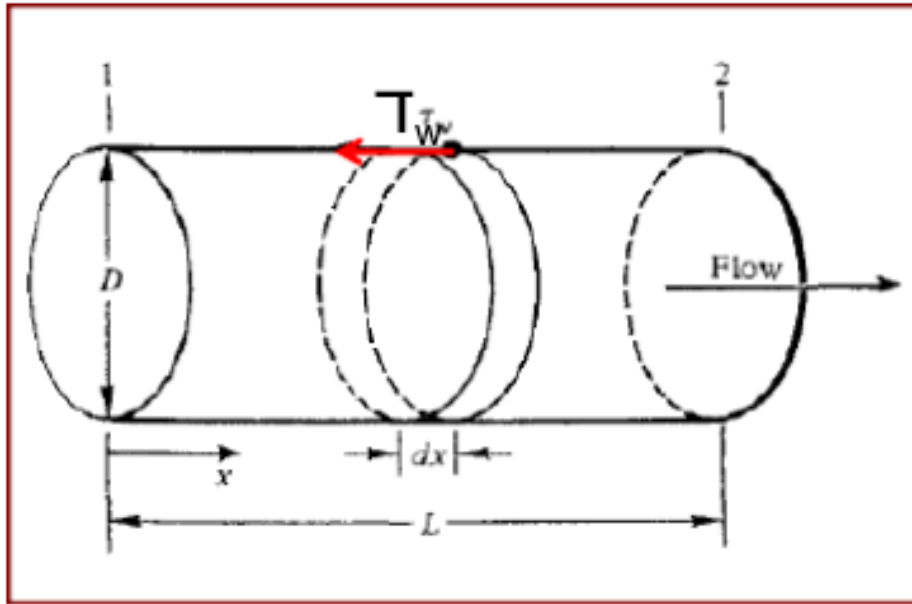
- Velocity change (due to friction) associated with entropy change
- Friction can only increase entropy
 - can only approach $M=1$
 - friction alone can not allow flow to transition between sub/supersonic -possible only through shock
- Two solutions given $(\rho v, h_o, s)$: subsonic & supersonic
 - change mass flux: new Fanno line



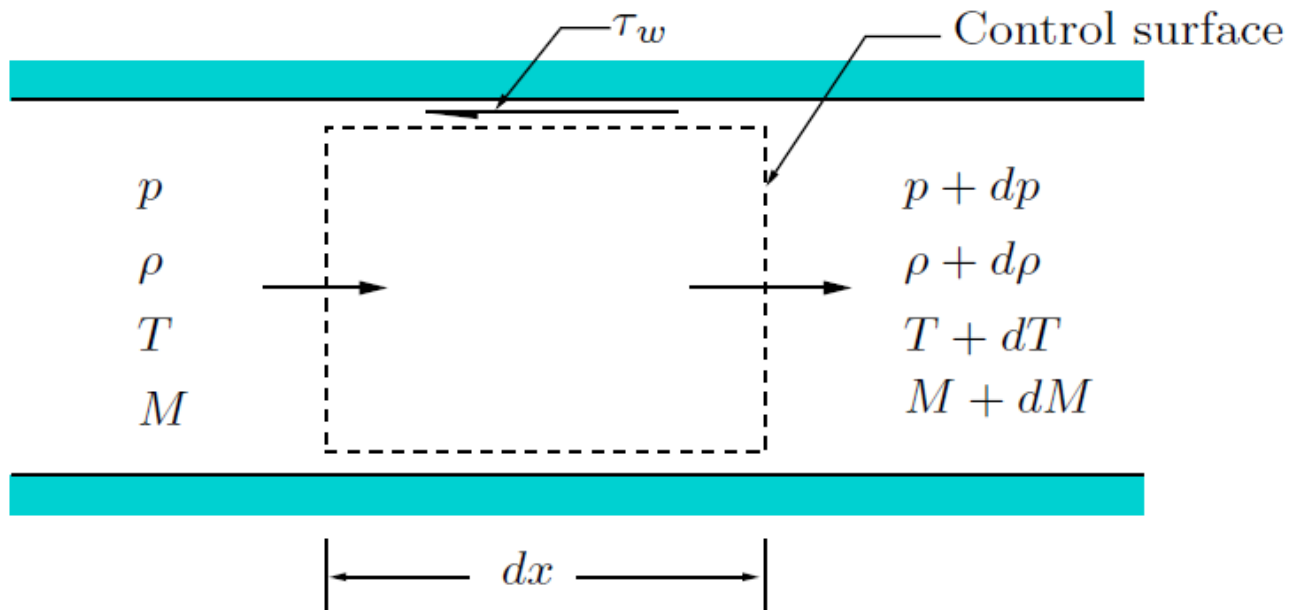
The effects of friction on the properties of Fanno flow

Property	Subsonic	Supersonic
Velocity, V	Increase	Decrease
Mach number, Ma	Increase	Decrease
Stagnation temperature, T_0	Constant	Constant
Temperature, T	Decrease	Increase
Density, ρ	Decrease	Increase
Stagnation pressure, P_0	Decrease	Decrease
Pressure, P	Decrease	Increase
Entropy, s	Increase	Increase

Analytical Analysis of Fanno Flow



Wall Shear Stress(τ_w) acts against the flow direction



Momentum balance for the small portion of the duct

Net pressure force - Force due to wall shear stress
= Mass flow rate · (Velocity out- Velocity in)

$$pA - (p + dp)A - \tau_w(\text{perimeter}) dx = \rho VA(V + dV - V)$$

$$-dp - \tau_w \frac{P}{A} dx = \rho V dV \quad -A dp - \tau_w dA_w = \dot{m} dV$$

Hydraulic diameter, D_H
$$D_H = \frac{4 (\text{Area})}{\text{Perimeter}} = \frac{4A}{P}$$

Dividing by ρV^2
$$-\frac{dp}{\rho V^2} - \frac{\tau_w}{\rho V^2} \frac{P}{A} dx = \frac{dV}{V}$$

$$D \equiv \frac{4A}{\left(\frac{dA_w}{dx}\right)} = 4 \frac{A}{dA_w} dx$$

$$-dp - 4f \frac{\rho V^2}{2} \frac{dx}{D} = \frac{\dot{m}}{A} dV = \rho V^2 \frac{dV}{V}$$

continuity of mass

$$\rho V = (\rho + d\rho)(V + dV)$$

Rearranging and ignoring small terms

$$\frac{dV}{V} + \frac{d\rho}{\rho} = 0$$

$$M^2 = \frac{V^2}{a^2} = \frac{V^2}{\gamma RT}$$

$$(M + dM)^2 = \frac{(V + dV)^2}{\gamma R(T + dT)}$$

$$\boxed{\frac{dM^2}{M^2} = \frac{dV^2}{V^2} - \frac{dT}{T}}$$

$$p = \rho RT$$

Eqn of State

$$p + dp = (\rho + d\rho)R(T + dT)$$

Definition of friction coefficient/Friction factor/Fanning friction factor, f

The *coefficient of drag*, or the *coefficient of friction*, as it is generally referred to for flow in ducts, is defined as

$$f \equiv \frac{\text{Wall shear stress}}{\text{Dynamic pressure head of the stream}}$$
$$f \equiv \frac{\tau_w}{\frac{1}{2}\rho V^2}$$

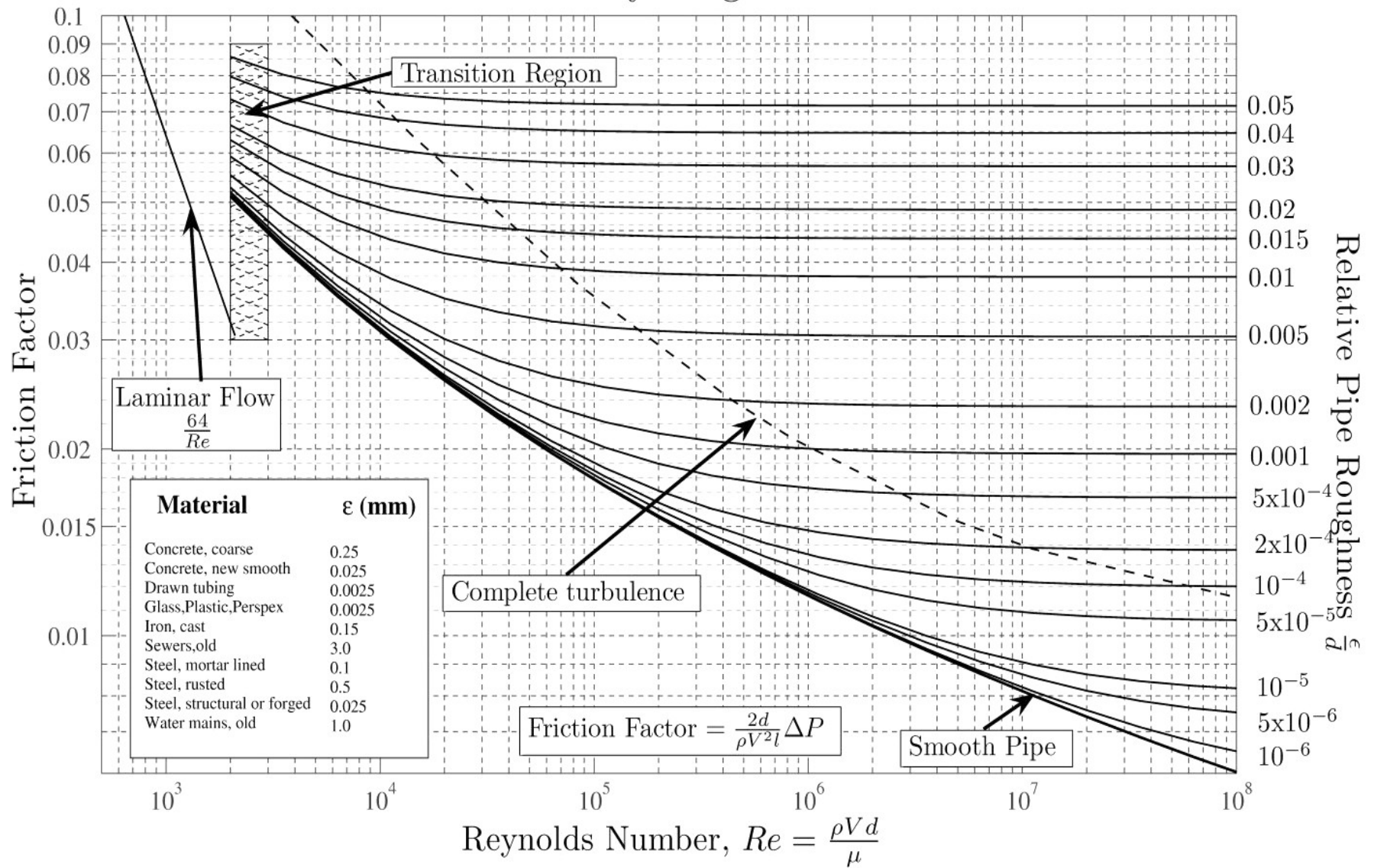
Darcy friction factor [f_d -Incompressible flow] $f = \frac{f_D}{4}$

ϵ is a measure of the mean height of the wall roughness

$$f = \text{function} (Re, \epsilon/D_H, M)$$

A Moody chart can be used to give the Darcy friction factor

Moody Diagram



From L.F. Moody, *Transactions of the ASME*, 671–684, Nov 1944. ASME

$$-dp - 4f \frac{\rho V^2}{2} \frac{dx}{D} = \frac{\dot{m}}{A} dV = \rho V^2 \frac{dV}{V}$$

$$\frac{dM^2}{M^2} = \frac{\gamma M^2 \left(1 + \frac{\gamma - 1}{2} M^2 \right)}{1 - M^2} 4f \frac{dx}{D}$$

$$\frac{dp}{p} = - \frac{\gamma M^2 \left(1 + (\gamma - 1) M^2 \right)}{2(1 - M^2)} 4f \frac{dx}{D}$$

$$\frac{dV}{V} = \frac{\gamma M^2}{2(1 - M^2)} 4f \frac{dx}{D}$$

$$\frac{dT}{T} = \frac{1}{2} \frac{da}{a} = - \frac{\gamma(\gamma - 1)M^4}{2(1 - M^2)} 4f \frac{dx}{D}$$

$$\frac{d\rho}{\rho} = - \frac{\gamma M^2}{2(1 - M^2)} 4f \frac{dx}{D}$$

$$\frac{dp_0}{p_0} = - \frac{\gamma M^2}{2} 4f \frac{dx}{D}$$

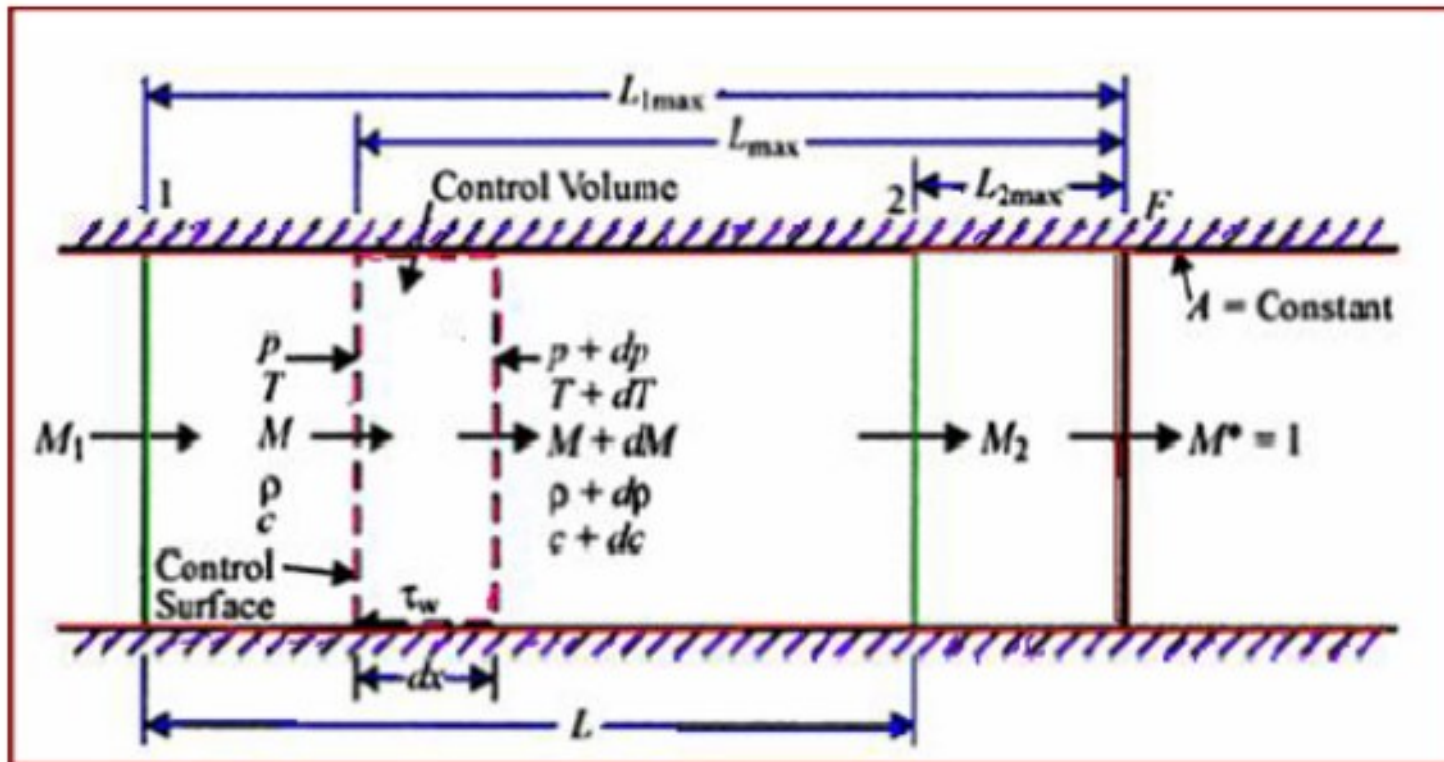
$$\frac{dF}{F} = - \frac{\gamma M^2}{2(1 + \gamma M^2)} 4f \frac{dx}{D}$$

$$\frac{ds}{c_p} = - \frac{\gamma - 1}{\gamma} \frac{dp_0}{p_0}$$

$$\frac{ds}{c_p} = \frac{(\gamma - 1) M^2}{2} 4f \frac{dx}{D}$$

'f' is positive (entropy must increase)- Tw opp to flow direction

Working relations-Fanno Flow



Working relations

$$\int_0^{L_{\max}} 4f \frac{dx}{D} = \int_{M^2}^1 \frac{1 - M^2}{\gamma M^4 \left(1 + \frac{\gamma - 1}{2} M^2 \right)} dM^2$$

$$4\bar{f} \frac{L_{\max}}{D} = \frac{1 - M^2}{\gamma M^2} + \frac{\gamma + 1}{2\gamma} \ln \left[\frac{(\gamma + 1)M^2}{2 \left(1 + \frac{\gamma - 1}{2} M^2 \right)} \right]$$

$\bar{f}$ is the mean friction coefficient

$$\bar{f} = \frac{1}{L_{\max}} \int_0^{L_{\max}} f dx$$

Duct length required for the flow to pass from initial Mach number M_1 to Final Mach number M_2

$$4\bar{f} \frac{L}{D} = \left(4\bar{f} \frac{L_{\max}}{D} \right)_{M_1} - \left(4\bar{f} \frac{L_{\max}}{D} \right)_{M_2}$$

Max. Condition, $M_2=1$

Local Flow properties

$$\frac{p}{p^*} = \frac{1}{M} \left[\frac{\gamma + 1}{2 \left(1 + \frac{\gamma - 1}{2} M^2 \right)} \right]^{1/2}$$

$$\frac{V}{V^*} = M \left[\frac{\gamma + 1}{2 \left(1 + \frac{\gamma - 1}{2} M^2 \right)} \right]^{1/2}$$

$$\frac{T}{T^*} = \frac{a^2}{a^{*2}} = \frac{\gamma + 1}{2 \left(1 + \frac{\gamma - 1}{2} M^2 \right)}$$

$$\frac{\rho}{\rho^*} = \frac{V^*}{V} = \frac{1}{M} \left[\frac{2 \left(1 + \frac{\gamma - 1}{2} M^2 \right)}{\gamma + 1} \right]^{1/2}$$

$$\frac{p_0}{p_0^*} = \frac{1}{M} \left[\frac{2 \left(1 + \frac{\gamma-1}{2} M^2 \right)}{\gamma+1} \right]^{(\gamma+1)/2(\gamma-1)}$$

$$\frac{F}{F^*} = \frac{1 + \gamma M^2}{M \left[2(\gamma+1) \left(1 + \frac{\gamma-1}{2} M^2 \right) \right]^{1/2}}$$

$$\frac{s - s^*}{c_p} = \ln M^2 \left[\frac{(\gamma+1)}{2M^2 \left(1 + \frac{\gamma-1}{2} M^2 \right)} \right]^{(\gamma+1)/(2\gamma)}$$

$$\frac{\rho_2}{\rho_1} = \frac{\left(\frac{\rho}{\rho^*} \right)_{M_2}}{\left(\frac{\rho}{\rho^*} \right)_{M_1}}$$

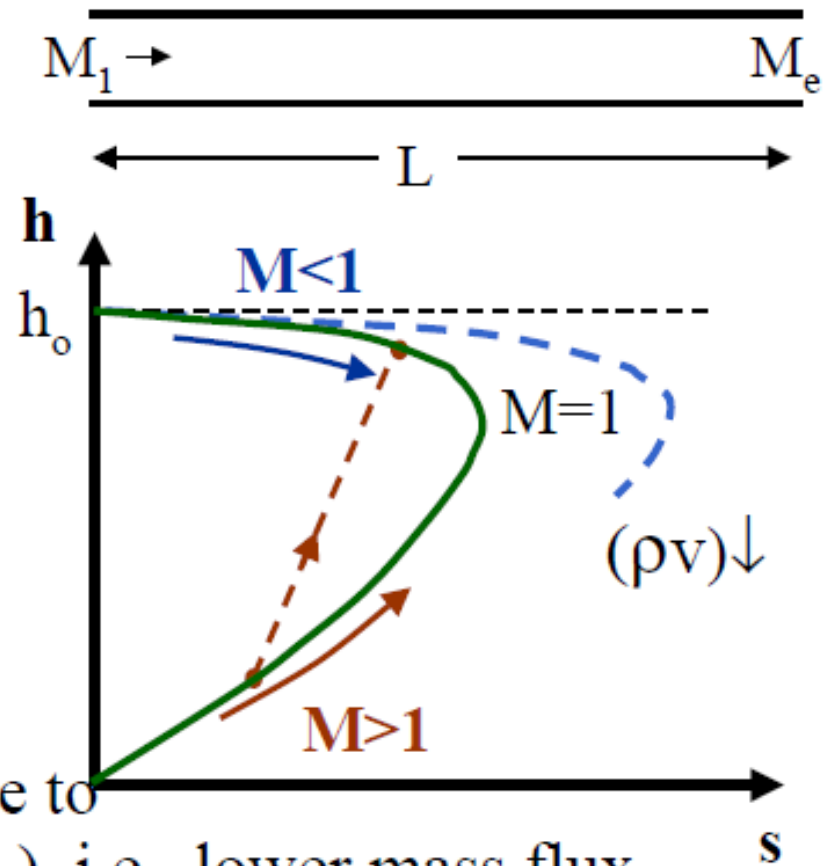
To find change in flow properties

Fanno Line - Choking

- Total friction experienced by flow increases with length of “flow”, e.g., duct length, L

- For long enough duct, $M_e = 1$ ($L = L_{\max}$)

- What happens if $L > L_{\max}$
 - flow already “choked”
 - **subsonic flow**: must move to different Fanno line (---), i.e., lower mass flux
 - **supersonic flow**: get a shock (---)

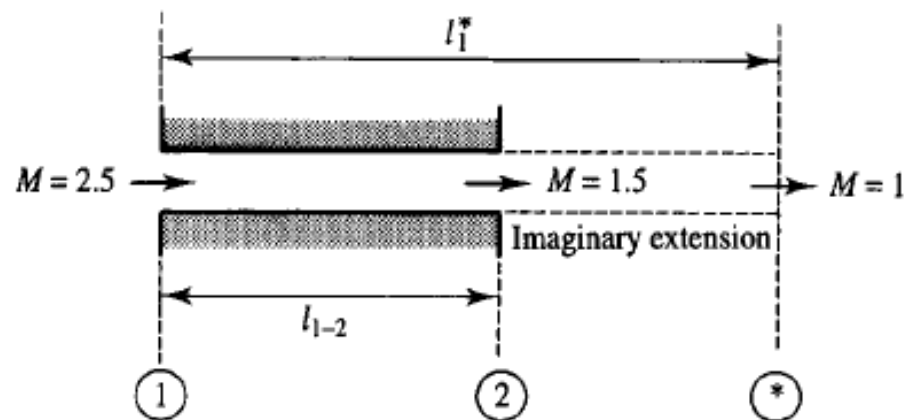


The Exit of supersonic nozzle connected to straight walled test section. Test section flows Air at $M_{\text{test}}=3.0$, $T_o=290.$ K, $p_o=500.$ kPa, $L=1\text{m}$, $D=10$ cm, $f=0.005$

Find:

- M , T , p at end of test section
- $p_{o,\text{exit}}/p_{o,\text{inlet}}$
- L_{max} for test section

Air flows in a 5 cm diameter pipe. The air enters at $M = 2.5$ and is to leave at $M = 1.5$. What length of pipe is required? What length of pipe would give $M = 1$ at the exit? Assume that $f = 0.002$ and that the flow is adiabatic.



$$\frac{4f l_{1-2}}{D} = \frac{4f l_1^*}{D} - \frac{4f l_2^*}{D} = 0.432 - 0.136$$

$$l_{1-2} = \frac{0.296 \times D}{4f} = \frac{0.296 \times 0.05}{4 \times 0.002} = 1.85 \text{ m}$$

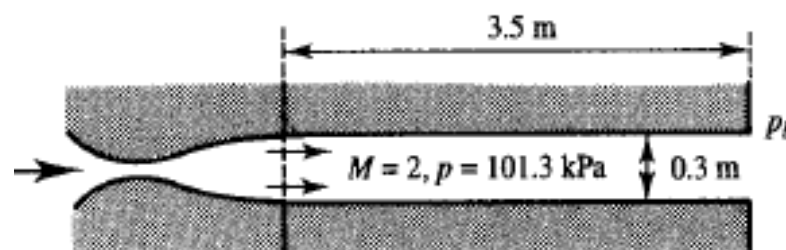
$$l_1^* = \frac{4f l_1^*}{D} \frac{D}{4f} = \frac{0.432 \times 0.05}{4 \times 0.002} = 2.7 \text{ m}$$

Air flows steadily from a large reservoir through a convergent-divergent nozzle into a 0.3 m diameter pipe with a length of 3.5 m. The conditions in the reservoir are such that the Mach number and the pressure at the inlet to the pipe are 2 and 101.3 kPa respectively. The average friction factor, f , for the flow in the pipe is estimated to be 0.005.

1. If no shocks occur, find M and p at the exit of the pipe
2. If there is a normal shock at the exit of the pipe, find the back pressure in the chamber into which the pipe is discharging
3. Find the back pressure in the chamber into which the pipe is discharging when there is a shock halfway down the pipe

1.

$$\begin{aligned}\frac{4fl_2^*}{D} &= \frac{4fl_1^*}{D} - \frac{4fl_{1-2}}{D} \\ &= 0.3049 - \frac{4 \times 0.005}{0.3} \times 3.5 = 0.0717\end{aligned}$$



Fanno table

$$p_2 = \frac{p_2}{p^*} \times \frac{p^*}{p_1} \times p_1 = \frac{0.715}{0.4083} \times 101.3 = 177.3 \text{ kPa}$$

2.

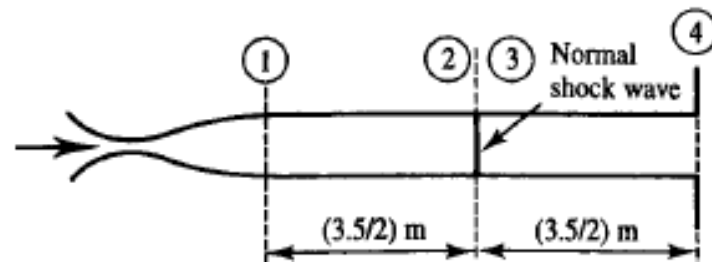
for $M = 1.32$, normal shock relations or tabl

$$\frac{p_2}{p_1} = 1.866$$

the back pressure is given by:

$$p_{\text{back}} = 1.866 \times 177.3 = 331 \text{ kPa}$$

3.



Fanno table for M_3

$$\frac{4f l_2^*}{D} = \frac{4f l_1^*}{D} - \frac{4f(l_1^* - l_2^*)}{D}$$

$$= 0.3049 - \frac{4 \times 0.005}{0.3} \left(\frac{3.5}{2} \right) = 0.19$$

$$\frac{4f l_3^*}{D} = 0.31, \quad \frac{p_3}{p^*} = 1.6$$

at the exit plane:

$$\begin{aligned} \frac{4f l_4^*}{D} &= \frac{4f l_3^*}{D} - \frac{4f(l_3 - l_4)}{D} \\ &= 0.31 - \left(\frac{4 \times 0.005}{0.3} \right) \left(\frac{3.5}{2} \right) = 0.194 \end{aligned}$$

Fanno table

$M_2 = 1.05$ and $p_2/p^* = 0.534$. Hence:

$M_4 = 0.71$ and $p_4/p^* = 1.48$. Hence:

$$p_2 = \frac{p_2}{p^*} \times \frac{p^*}{p_1} \times p_1 = \frac{0.534}{0.408} \times 101.3 = 132.6 \text{ kPa}$$

$$p_4 = p_{\text{back}} = \frac{p_4}{p^*} \times \frac{p^*}{p_3} \times \frac{p_3}{p_2} \times p_2$$

Normal shock table

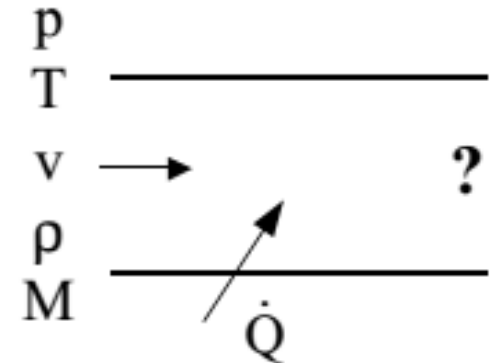
$$= \frac{1.48}{1.6} \times 3.01 \times 132.6 = 369 \text{ kPa}$$

$$\frac{p_3}{p_2} = 3.010, \quad M_3 = 0.654$$

Rayleigh Flow - Thermodynamics

- simple T_0 change

- Steady, 1-d, constant area, inviscid flow with no external work but *with reversible heat transfer (heating or cooling)*



- Conserved quantities (mass, momentum eqs.)
 - since $A=\text{const}$: mass flux= $\rho v=\text{constant}\equiv G$
 - since no forces but pressure: $p+\rho v^2=\text{constant}$
 - combining: $p+G^2/\rho=\text{constant}$
- On h - s diagram, can draw this **Rayleigh Line**

Governing Equations

For flow of gas through a constant-area duct without friction, the momentum equation may be written as

$$p + \rho V^2 = \frac{F}{A} = \text{constant}$$

By continuity,

$$\rho V = \frac{\dot{m}}{A} = G = \text{constant}$$

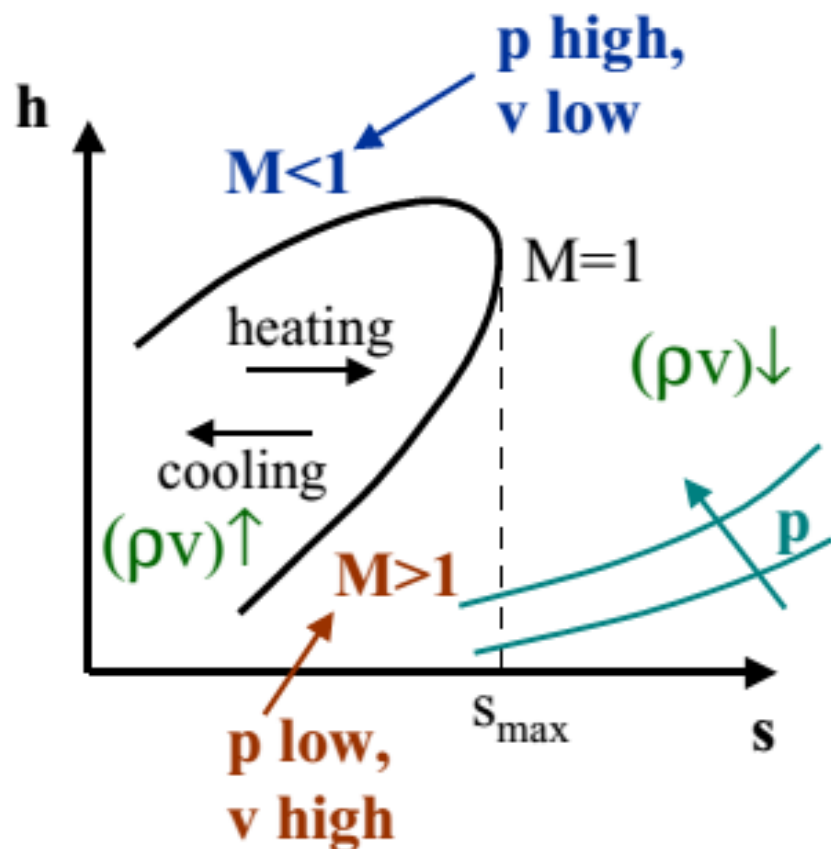
Combining Eqs.

$$p + \frac{G^2}{\rho} = \frac{F}{A}$$

For constant values of G and F/A , defines a unique relation between pressure and density, called ***Rayleigh line***. Since both the enthalpy h and entropy s are functions of p and ρ , eqn be used for representing Rayleigh line on the h - s diagram, as illustrated

Rayleigh Line

- $p + \rho v^2 = \text{constant}$,
 $\Rightarrow$ high p results in low v
- $ds = \delta Q/T$ (reversible)
 - heating increases s
 - cooling decreases s
 - $M=1$ at maximum s
- Change mass flux
 new Rayleigh line

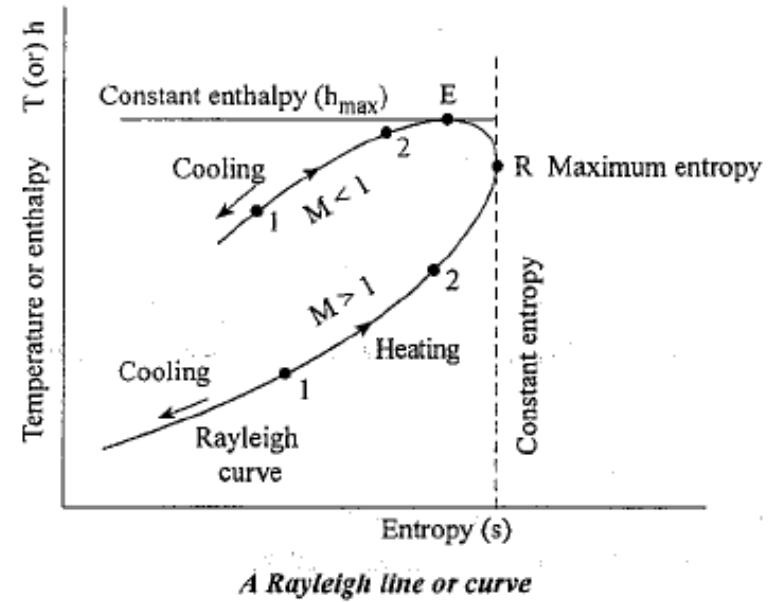
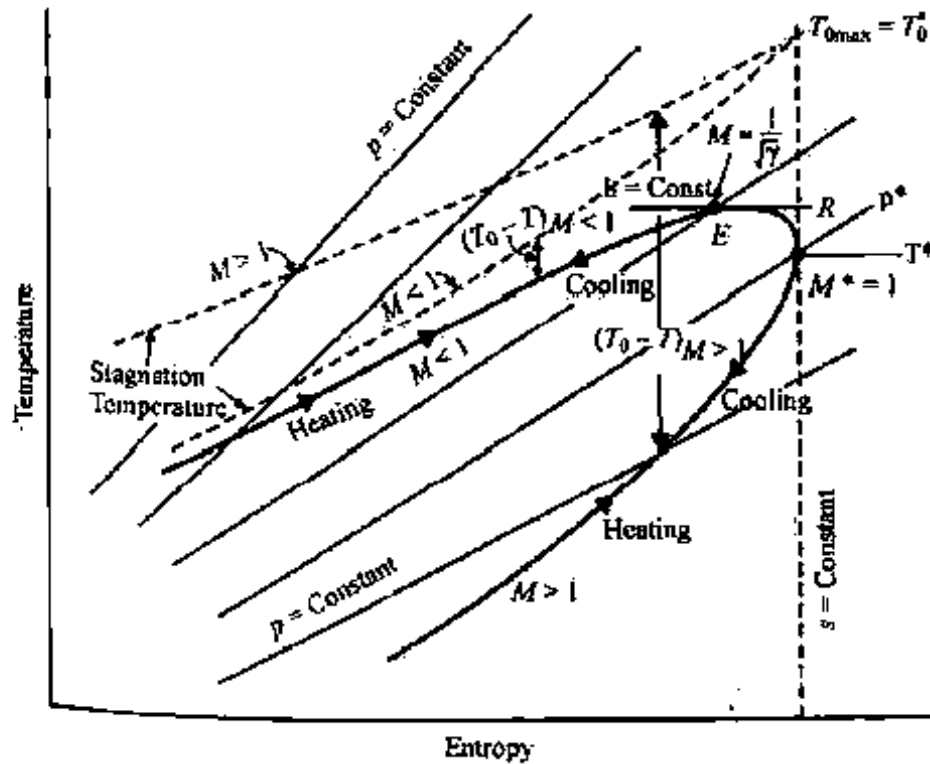


Rayleigh Flow Curve

- The portion above maximum entropy on the Rayleigh curve → subsonic flow
- The portion below maximum entropy on the Rayleigh curve → supersonic flow
- Heat addition is a reversible process → causes entropy increase
- Heat rejection → causes entropy decrease
- At subsonic speeds:
 - Heating → Mach number increases
 - Cooling → Mach number decreases
- At supersonic speeds:
 - Heating → Mach number decreases
 - Cooling → Mach number increases
- Heat addition always drives Mach number toward 1
- Cooling always drives Mach number away from 1

Rayleigh Flow Curve

$$T_0 = T_0^* = T_{0 \max}$$



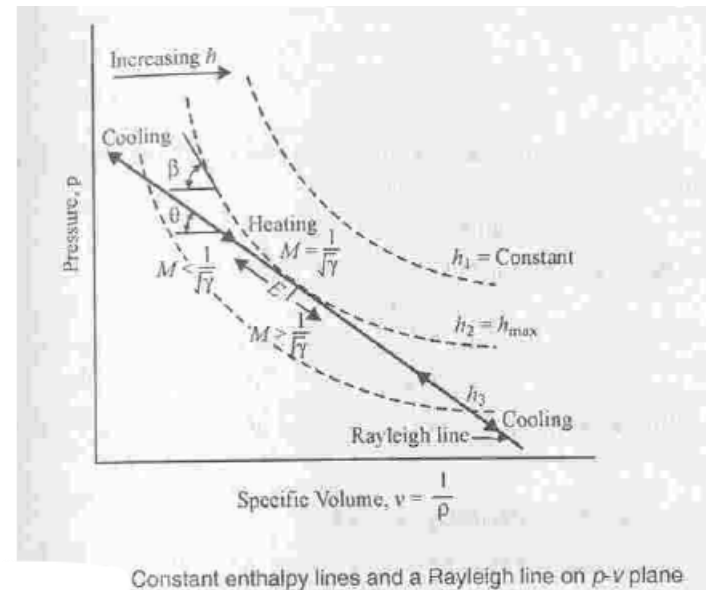
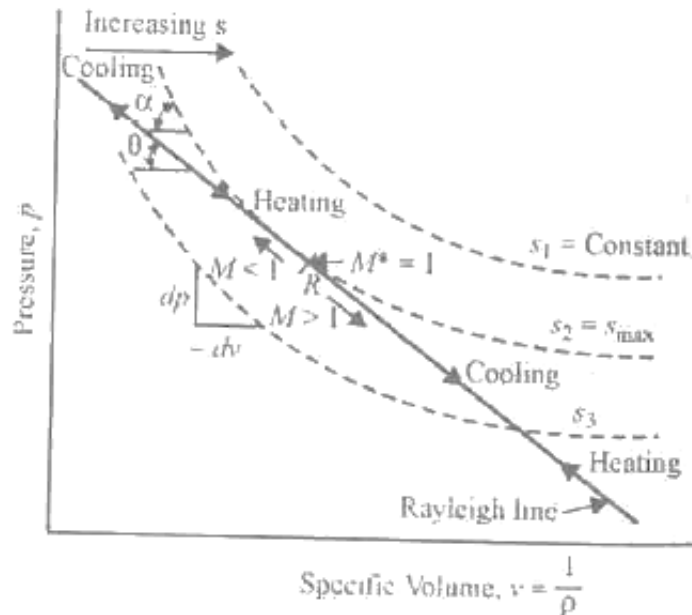
Slope of Rayleigh Line

two state points along the Rayleigh line gives:

$$p_1 + G^2 v_1 = p_2 + G^2 v_2$$

$$\frac{p_1 - p_2}{v_1 - v_2} = -G^2 = -\left(\frac{m}{A}\right)^2$$

$$\left(\frac{dp}{dv}\right)_R = -G^2 = \tan \theta = -(\rho e)^2 = -\rho^2 a^2 M^2$$



Slope of const Entropy lines

$$\tan \alpha = \frac{dp}{dv} = \frac{dp}{d\left(\frac{1}{\rho}\right)} = -\rho^2 \frac{dp}{d\rho}$$

But at constant entropy

$$dp = (dp/d\rho)_s * d\rho = a^2$$

Therefore,

$$\tan \alpha = dp/dv = -\rho^2 a^2$$

At the limiting state point R the slopes of the Rayleigh and the constant entropy line are same:

$$\tan \theta = \tan \alpha$$

Therefore

$$\rho^2 a^2 M^2 = \rho^2 a^2$$

$$M = M' = 1$$

This shows that the velocity of gas at the maximum entropy point is sonic.

Slope of const Enthalpy lines

$$Pv = RT = \text{constant}$$

Therefore, the slope of the constant enthalpy line is

$$\tan \beta = (dp/dv)_h = -p/v = -\rho p$$

The velocity of sound is given by

$$a^2 = \gamma p / \rho$$

$$p = \rho a^2 / \gamma$$

Therefore,

$$\tan \beta = (dp/dv)_h = -\rho^2 a^2 / \gamma$$

At the maximum enthalpy point E the slopes of the Rayleigh line and the constant enthalpy line are same,

$$\tan \theta = \tan \beta$$

Therefore

$$\rho^2 a^2 M^2 = \rho^2 a^2 / \gamma$$

$$M = M^* = 1/\sqrt{\gamma}$$

Rayleigh Flow

Energy Conservation

$$dQ = \dot{m} \left[c_p(T + dT) + \frac{(V + dV)^2}{2} - c_p T - \frac{V^2}{2} \right]$$

$$dq = c_p dT + V dV$$

$$\frac{dq}{dV} = c_p \frac{T}{V} - \frac{V}{\gamma - 1}$$

- At low velocities, the first term dominates
→ $dq/dV > 0$

- Adding heat increases velocity
- Removing heat decreases velocity

- At high velocities, the second term dominates → $dq/dV < 0$

- Adding heat decreases velocity
- Removing heat increases velocity

- Transition point occurs when $dq/dV = 0$

$$\frac{ds}{c_p} = \frac{dT}{T} - \frac{(\gamma - 1)}{\gamma} \frac{dp}{p}$$

$$\frac{ds}{c_p} = (1 - M^2) \frac{dV}{V}$$

$dVdq > 0$ when $M < 1$; $dq/dV < 0$ when $M > 1$

- Heat addition → always increases entropy
- Heat removal → always decreases entropy
- Entropy is maximized at Mach number $M=1$
- Heat drives the flow's entropy toward $M=1$

Rayleigh Flow- Max Heat transfer

$$\text{Heat transfer, } Q = mc_p (T_{02} - T_{01})$$

$$\Rightarrow \frac{Q}{c_p T_1} = \frac{T_{01}}{T_1} \left(\frac{T_{02}}{T_{01}} - 1 \right)$$

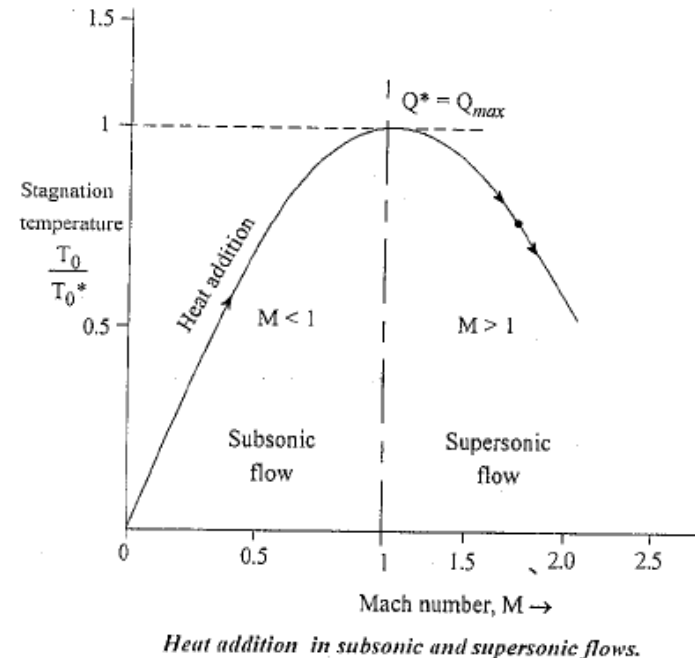
$$\frac{Q}{c_p T_1} = \frac{M_2^2 - M_1^2}{2M_1^2 (1 + \gamma M_2^2)^2} \left[2(1 - \gamma M_1^2 M_2^2) + (\gamma - 1)(M_2^2 + M_1^2) \right]$$

$$\frac{Q^*}{c_p T} = \frac{1 - M^2}{2M^2 (1 + \gamma)^2} \left[2(1 - \gamma M^2) + (\gamma - 1)(1 + M^2) \right]$$

$$\Rightarrow \frac{Q^*}{c_p T} = \frac{(1 - M^2)^2}{2M^2 (1 + \gamma)}$$

$$\frac{Q_{max}}{c_p T} = \frac{(1 - M^2)^2}{2(1 + \gamma)M^2}$$

$$\Rightarrow Q_{max} = \frac{(1 - M^2)^2}{2(1 + \gamma)M^2} \times c_p T$$

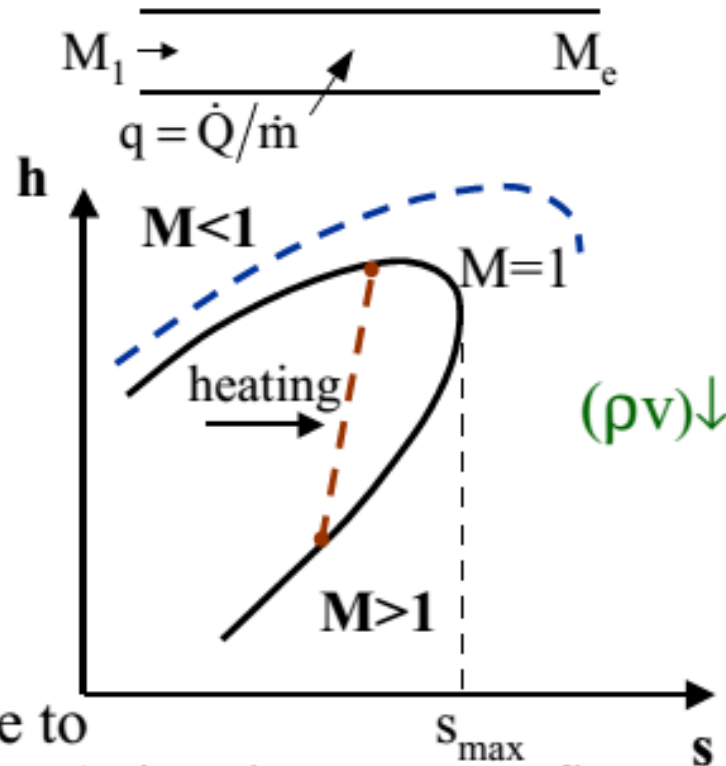


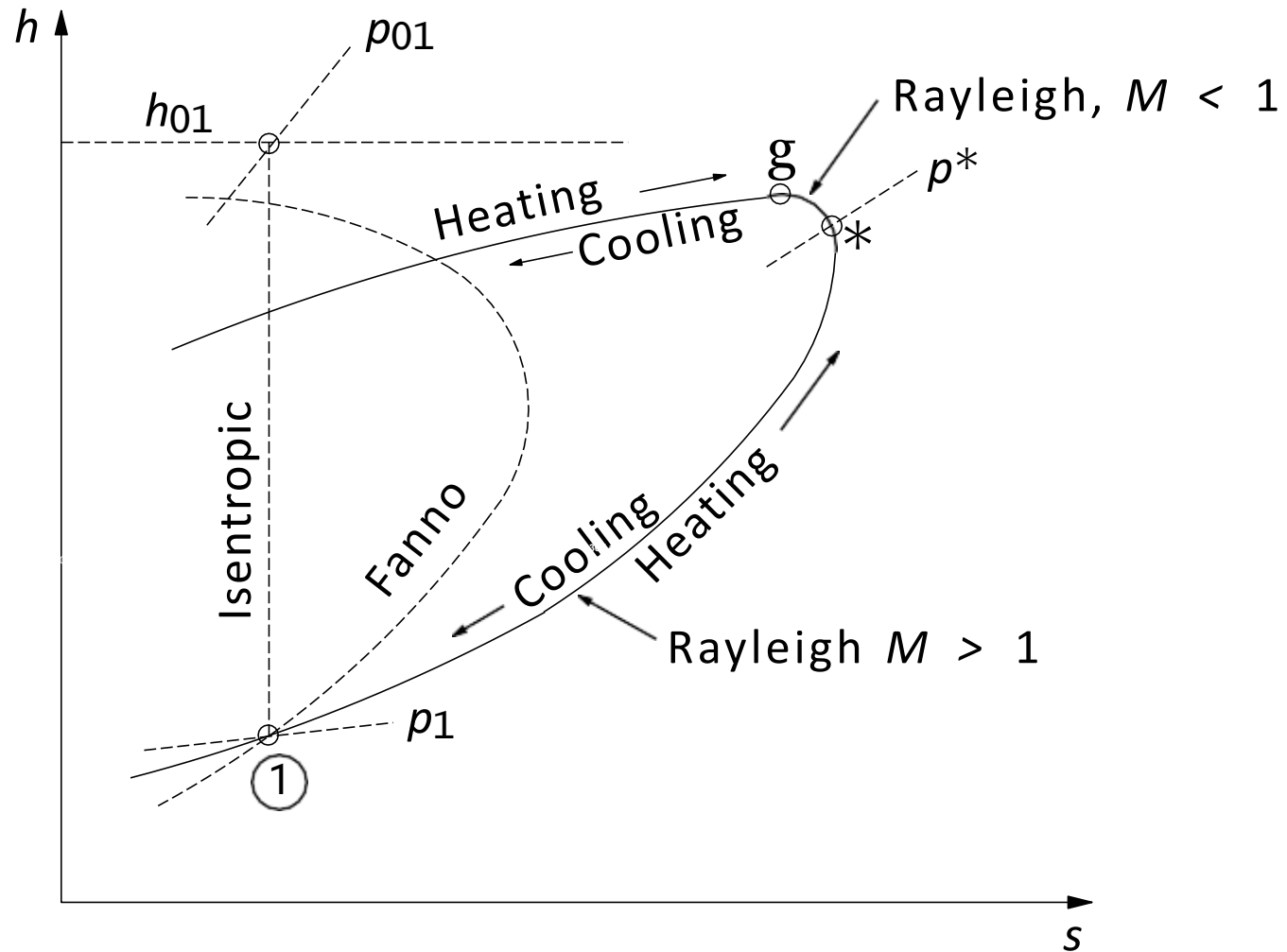
Heat addition in subsonic and supersonic flows.

Rayleigh Line - Choking

- Heat addition can only increase entropy
- For **enough heating**, $M_e = 1$ ($Q_{in} = Q_{max}$)
 - heat addition can lead to choked flow

- What happens if $Q_{in} > Q_{max}$
 - subsonic flow**: must move to different Rayleigh line (---), i.e., lower mass flux
 - supersonic flow**: get a shock (---)
 - stays on same Rayleigh line: ρv and $p + \rho v^2$ also const. for normal shock





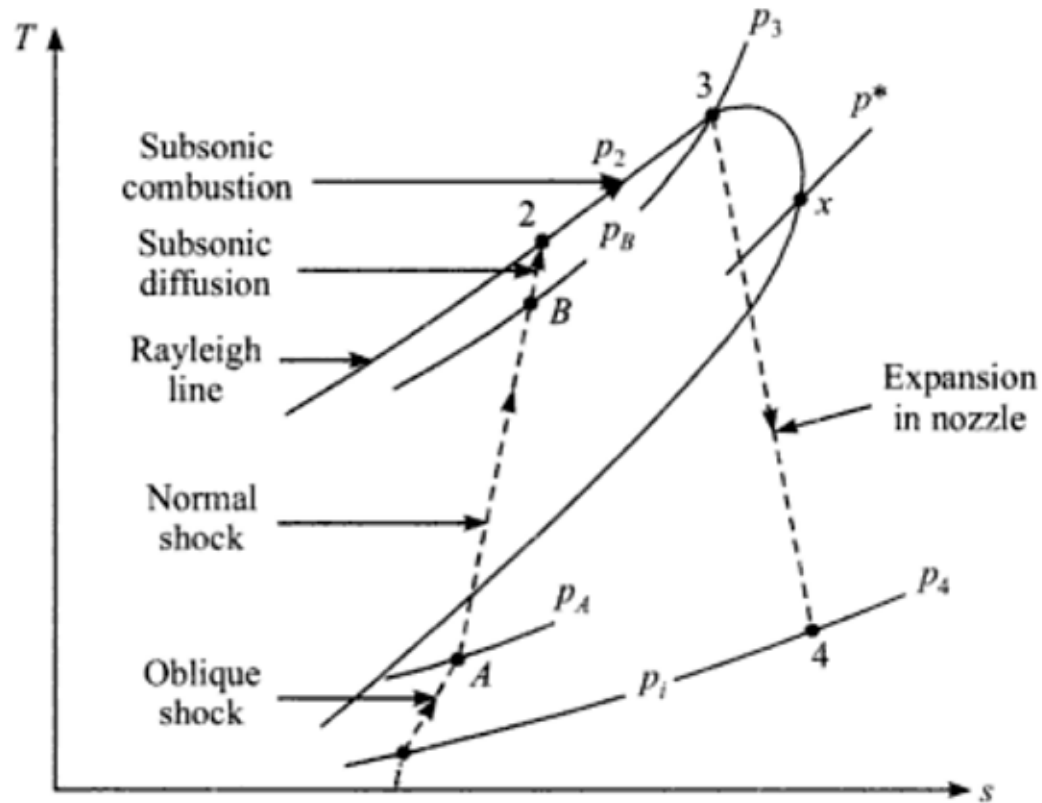
Rayleigh curve for simple T_0 -change.

	<u>Heating</u> <u>$M < 1$</u>	<u>Heating</u> <u>$M > 1$</u>	<u>Cooling</u> <u>$M < 1$</u>	<u>Cooling</u> <u>$M > 1$</u>
T_0	increases	increases	decreases	decreases
M	increases	decreases	decreases	increases
T	Note (1)	increases	Note (2)	decreases
p	decreases	increases	increases	decreases
p_0	decreases	decreases	increases	increases
V	increases	decreases	decreases	increases

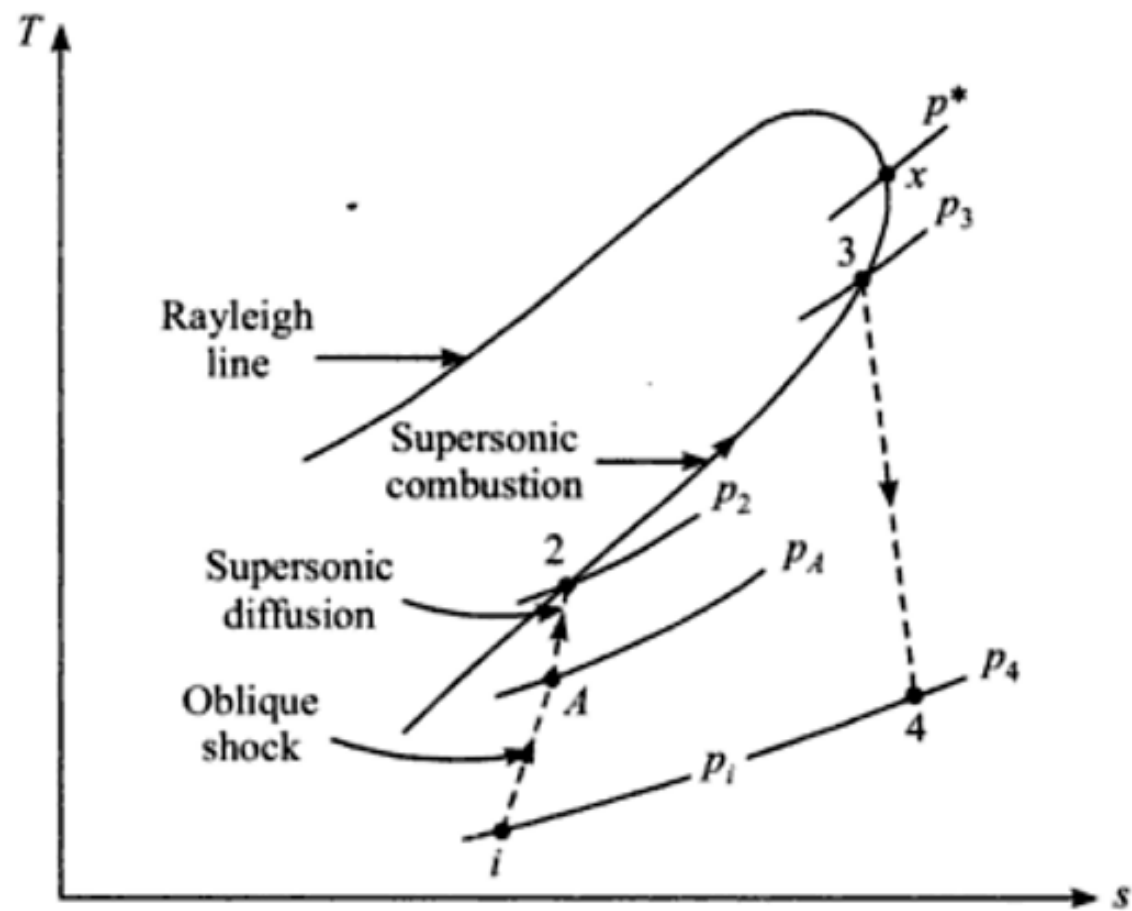
Note (1): increases for $M < 1/\sqrt{k}$, and decreases for $M > 1/\sqrt{k}$

Note (2): decreases for $M < 1/\sqrt{k}$, and increases for $M > 1/\sqrt{k}$

Application of Rayleigh flow

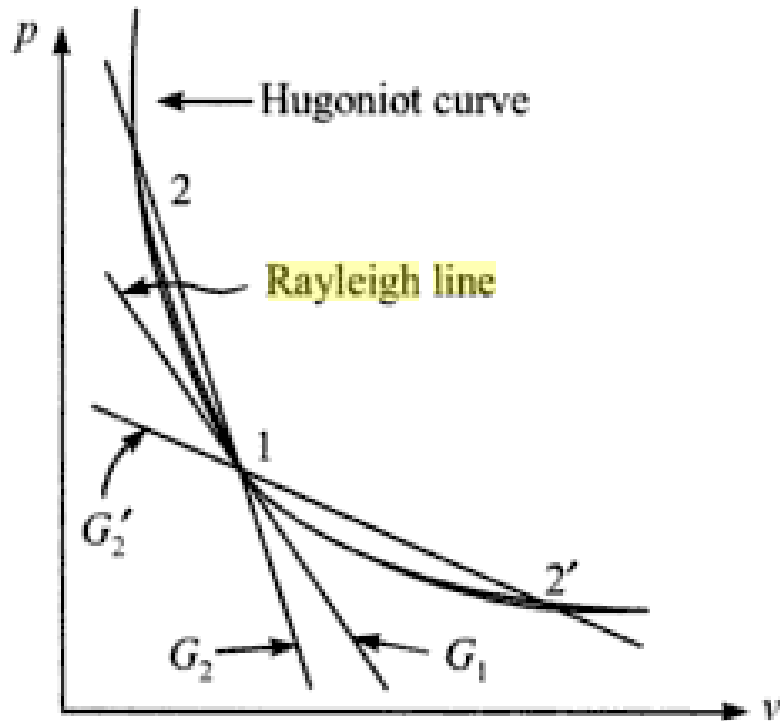


T-s diagram of ramjet engine process.



$T-s$ diagram of scramjet engine process.

Shock waves and combustion waves in a detonation or deflagration process



- **Point 1:** Initial state (pre-shock or pre-combustion).
- **Point 2 and 2':** Possible final states post-shock (or post detonation/deflagration), where 2 is likely a **detonation point** (high pressure, low volume) and 2' is a **deflagration point** (lower pressure, higher volume).

Mass Flux Lines G_1, G_2, G_2' :

- Different Rayleigh lines for different mass fluxes.
- G_1 corresponds to the initial Rayleigh line from state 1.
- G_2, G_2' show how changing the mass flux shifts the Rayleigh line (steeper or flatter).

- The **intersection of the Rayleigh line and Hugoniot curve** determines the physically possible states after a combustion or shock process.
- The shape of the Hugoniot curve and the slope of the Rayleigh line determine whether the process is: Strong detonation (steep Rayleigh line intersecting upper Hugoniot branch), Weak deflagration, or Impossible (if no intersection occurs).

Problem

Air flows through a constant area duct. The pressure and temperature of the air at the inlet to the duct are 100 kPa and 10°C respectively and the inlet Mach number is 2.8. Heat is transferred to the air as it flows through the duct and as a result the Mach number at the exit is 1.3.

Find the pressure and temperature at the exit. If no shock waves occur in the flow, find the maximum amount of heat that can be transferred to the air per unit mass of air. Also find the exit pressure and temperature that would exist with this maximum heat transfer rate. Assume that the flow is steady, that the effects of wall friction can be neglected and that the air behaves as a perfect gas. Assume $C_p = 1.007 \text{ kJ/kg}^\circ\text{K}$

isentropic tables gives $T_{01}/T_1 = 2.568$.

$$T_{01} = 2.568 \times (273 + 10) = 726.7 \text{ K}$$



Rayleigh flow table

For $M = 2.8$:

$$\frac{p}{p^*} = 0.2004, \quad \frac{T}{T^*} = 0.3149, \quad \frac{T_0}{T_0^*} = 0.6738$$

For $M = 1.3$:

$$\frac{p}{p^*} = 0.7130, \quad \frac{T}{T^*} = 0.8592, \quad \frac{T_0}{T_0^*} = 0.9580$$

$$p_2 = \frac{p_2/p^*}{p_1/p^*} p_1 = \frac{0.7130}{0.2004} \times 100 = 355.8 \text{ kPa}$$

$$T_2 = \frac{T_2/T^*}{T_1/T^*} T_1 = \frac{0.8592}{0.3149} \times 283 = 772.2 \text{ K} = 499.2^\circ\text{C}$$

at the exit is 1 and $T_{02} = T_0^*$. Hence, in this case:

$$T_{02} = \frac{T_{01}}{T_{01}/T_0^*} = \frac{726.7}{0.6738} = 1078.5 \text{ K}$$

But: $q = c_p(T_{02} - T_{01}) = 1.007(1078.5 - 726.7) = 354.3 \text{ kJ/kg}$

$$p_2 = p^* = \frac{p_1}{p_1/p^*} = \frac{100.7}{0.2004} = 499.0 \text{ kPa}$$

$$T_2 = T^* = \frac{T_1}{T_1/T^*} = \frac{283}{0.3149} = 898.7 \text{ K} = 625.7^\circ\text{C}$$

The data for a gas ($\gamma = 1.3$, $c_p = 2.144 \text{ kJ/kg K}$) at entry of combustion chamber are $c_1 = 150 \text{ m/s}$, $p_1 = 4 \text{ bar}$ and $T_1 = 395 \text{ K}$. If the exit Mach number is 0.78, calculate the following

1. The initial Mach number
2. Final pressure, temperature and velocity of the gas
3. Stagnation pressure loss
4. Air fuel ratio required.

Take calorific value of fuel is 42 MJ/kg

Result:

1. $M_1 = 0.297$
2. $p_2 = 2.44 \times 10^5 \text{ N/m}^2$
 $T_2 = 1037.13 \text{ K}$
 $c_2 = 633.24 \text{ m/s}$
3. $\Delta p_0 = 0.6 \times 10^5 \text{ N/m}^2$
4. Air fuel ratio = 25.70 : 1